



S.B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT & RESEARCH, NAGPUR

Practical 10

Aim: Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

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Semester / Year: 4th / 2nd

Academic Year: 2025-2026

Date of Performance:

Date of Submission:

❖ CODE :



```
GNU nano 8.7
#include <stdio.h>
#include <stdlib.h>

int main() {

    int request[] = {98,183,41,122,14,124,65,67};
    int visited[8] = {0};

    int head = 53;
    int total = 0;

    for(int i=0;i<8;i++) {

        int min = 1000;
        int index = -1;

        for(int j=0;j<8;j++) {

            if(visited[j]==0) {

                int distance = abs(head - request[j]);

                if(distance < min) {
                    min = distance;
                    index = j;
                }

            }

        }

        visited[index] = 1;
        total += min;

        head = request[index];

        printf("Head moves to cylinder %d\n", head);
    }

    printf("\nTotal Head Movement = %d\n", total);

    return 0;
}
```

Help
 Exit

Write Out
 Read File

Where Is
 Replace

Cut
 Paste

Execute
 Justify

❖ OUTPUT :

 ~

```
De11@DESKTOP-D7MT61A UCRT64 ~  
$ nano sstf.c
```

```
De11@DESKTOP-D7MT61A UCRT64 ~  
$ gcc sstf.c -o sstf.exe
```

```
De11@DESKTOP-D7MT61A UCRT64 ~  
$ ./sstf.exe
```

```
Head moves to cylinder 41  
Head moves to cylinder 65  
Head moves to cylinder 67  
Head moves to cylinder 98  
Head moves to cylinder 122  
Head moves to cylinder 124  
Head moves to cylinder 183  
Head moves to cylinder 14
```

```
Total Head Movement = 323
```

```
De11@DESKTOP-D7MT61A UCRT64 ~  
$ |
```